Unit F1 Further Pure Mathematics 1

IAS compulsory unit for Further Mathematics and Pure Mathematics — Externally assessed

F1.1 Unit description

Complex numbers; roots of quadratic equations; numerical solution of equations; coordinate systems; matrix algebra; transformations using matrices; series; proof.

F1.2 Assessment information

Prerequisites

A knowledge of the specification for C12, its prerequisites, preambles and associated formulae, is assumed and may be tested.

It is also necessary for students:

- to have a knowledge of location of roots of $f(x) = 0$ by considering changes of sign of $f(x)$ in an interval in which $f(x)$ is continuous
- to have a knowledge of rotating shapes through any angle about (0, 0)
- to be able to divide a cubic polynomial by a quadratic polynomial
- to be able to divide a quartic polynomial by a quadratic polynomial.

Examination

The examination will consist of one $1\frac{1}{2}$ hour paper. It will contain about nine questions of varying length. The mark allocations per question will be stated on the paper. All questions should be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π, x^2, $\sqrt{x}$, $\frac{1}{x}$, x^y, $\ln x$, e^x, $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Formulae which students are expected to know are given overleaf and these will not appear in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.